

Executive Summary: Solving India's Regulatory & Quality Compliance Bottleneck
India enforces over **19,000+ Bureau of Indian Standards (BIS)** documents and dozens of expanding Central **Quality Control Orders (QCOs)**. Non-compliance leads to factory seals, criminal penalties under Section 29 of the BIS Act 2016, and blocked imports. However, MSMEs and manufacturers lack accessible legal-technical expertise. Generic LLMs (like ChatGPT) fail catastrophically in this domain because they **hallucinate non-existent standard numbers** and cannot track mandatory legal QCOs. **Manak AI solves this with a zero-hallucination Dual Defense Engine: (1) Mathematical Retrieval Grounding, and (2) Automated Compliance Gap Analysis.**

CORE INNOVATION: WHY GENERIC AI FAILS VS HOW MANAK AI SOLVES IT

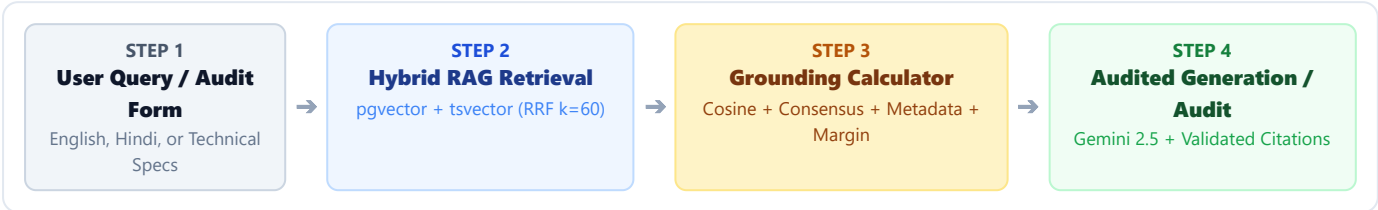
✗ The Problem with Generic LLMs (ChatGPT / Claude)

- Confidently Hallucinates:** Fabricates fictitious IS codes, mixing up ISO with Indian Standards.
- No Legal QCO Awareness:** Cannot tell whether a standard is voluntary or statutory mandatory under Gazette notifications.
- Zero Verifiability:** Provides free-form text without clause, table, or page-level citations.
- Dangerous for MSMEs:** Misleading advice leads to impounded inventory and BIS seizure notices.

✓ The Manak AI Solution (Dual-Engine Architecture)

- Deterministic Retrieval Grounding:** A 4-factor mathematical scoring system validates every retrieval before generation.
- Strict Refusal Policy:** If grounding is below 38%, the system refuses to guess—protecting users from bad advice.
- Clause-Level Legal Citations:** Every assertion maps to exact Standard Number, Clause, and Page Number (e.g., *IS 14543:2016 Cl. 4.2*).
- Interactive Gap Analysis:** Compares declared manufacturer tests against statutory BIS STI checklists.

SYSTEM ARCHITECTURE: FROM QUERY TO GROUNDED COMPLIANCE



VERIFIED BENCHMARK DEMO CATEGORIES

PRODUCT CATEGORY	APPLICABLE IS STANDARD	MANDATORY QCO AUTHORITY	CRITICAL HIGH-RISK TEST VERIFIED
Stainless Steel Bottles	IS 14543:2016	DPIIT Mandatory QCO	Grade 304/316 Food Contact & 2-Hr Boiling Citric Acid Immersion
Domestic Pressure Cookers	IS 2347:2017	DPIIT Mandatory QCO	Dual Safety Relief Valves (1.4-1.8x) & Proof Burst Test (3.0x)
Packaged Drinking Water	IS 10500:2012 / IS 14543	Ministry of Health & BIS	Microbiological Pathogen Colony Count & Toxic Heavy Metals Scan
Toys (Mechanical & Chemical)	IS 9873:Part 1-3	DPIIT Toys QCO (2020)	Small Parts 31.7mm Cylinder Choking Hazard & Heavy Metal Migration

DECONSTRUCTING THE JUDGE UI: WHAT DOES "83% HIGH GROUNDING" MEAN?

Retrieval Grounding Score

83% HIGH Grounding

"Authoritative match found in IS 10500:2012. Multiple verified clauses retrieved with strong consensus across keyword and semantic indices."

Vector Semantic Cosine:	93.0%
Hybrid RRF Consensus:	100.0%
Exact Metadata Match:	100.0%

What Judges Must Notice

Most AI apps ask an LLM "How confident are you from 1 to 100?"—which is arbitrary and hallucinated.

Manak AI is fundamentally different: The confidence score is computed **before the LLM writes a single word**. It is a pure mathematical function of retrieval precision across PostgreSQL vector embeddings, PostgreSQL tsvector keyword rank, and gazette metadata.

THE 4 MATHEMATICAL SIGNALS OF THE GROUNDING ENGINE

Weighted Score = (0.35 × TopSimilarity) + (0.25 × HybridConsensus) + (0.25 × MetadataMatch) + (0.15 × ScoreMargin)

1. Vector Semantic Cosine (Weight: 35%)

93.0% in Demo

Measures the high-dimensional cosine similarity between the Gemini embedding vector of the user's natural language query and the indexed text chunks in Supabase pgvector. Ensures conceptual understanding even if terminology varies.

2. Hybrid RRF Consensus (Weight: 25%)

100.0% in Demo

Calculates the intersection ratio between dense vector search and sparse full-text keyword search (tsvector). A 100% consensus indicates that the exact same clauses were ranked top-tier by both semantic meaning AND exact keyword frequency.

3. Exact Metadata Match (Weight: 25%)

100.0% in Demo

Confirms deterministic match of the standard number (e.g., "IS 10500" or "IS 14543"), edition year, and product classification in the database schema. Eliminates false associations between unrelated products.

4. Score Margin / Separation (Weight: 15%)

Rank Separation

Measures the mathematical delta between the 1st and 5th retrieved results in Reciprocal Rank Fusion (topScore - fifthScore). A high margin proves clear retrieval convergence rather than noisy, ambiguous matches.

THRESHOLD CLASSIFICATION & THE "REFUSAL" SAFETY GUARANTEE

SCORE TIER	VISUAL BADGE	SYSTEM BEHAVIOR	COMPLIANCE ACTION
Score ≥ 65%	HIGH Grounding	Authoritative match found in gazette standard. Full answer generated with active clause citations.	Directly actionable for audit preparation and QCO compliance.
38% ≤ Score < 65%	MEDIUM Grounding	General clauses found, but sub-clauses lack exact consensus. Answer includes cautionary flags.	Manufacturer advised to cross-check with designated BIS technical officer.
Score < 38%	LOW Grounding	REFUSAL TRIGGER: System explicitly states no authoritative evidence exists. Does NOT hallucinate.	Protects manufacturer from following unverified guidance.

WHAT IS THE COMPLIANCE GAP ANALYSIS TOOL?

The **Compliance Gap Analysis & Audit Tool** functions as a digital pre-audit inspection officer. Instead of paying lakhs of rupees to compliance consultants, an MSME enters their product materials, manufacturing process, declared tests, and certifications. The engine executes a clause-by-clause comparison against the official **Scheme of Testing and Inspection (STI)** mandated by BIS.

THE 5 CORE PILLARS EVALUATED DURING EVERY AUDIT

1. Raw Material Compliance

Verifies input metallurgy/chemistry. E.g., for utensils/bottles, strictly checks for Grade 304/316 austenitic stainless steel per IS 6911; flags illegal 200-series grades.

2. Mandatory Testing Protocols

Cross-references manufacturer's routine tests against mandatory statutory tests (e.g. 2-hr boiling citric acid, hydrostatic burst, small parts safety cylinder).

3. Regulatory QCO / Scheme-I

Catches the #1 industry misconception: having an *ISO 9001 certificate* does NOT substitute for statutory BIS ISI certification under Central QCOs.

4. Manufacturing Infrastructure & STI

Audits factory readiness: requires an in-house quality control testing laboratory with calibrated instruments, qualified test personnel, and batch records.

5. Product Marking & Traceability

Inspects indelible laser-etching or embossing requirements: ISI monogram, 7-10 digit CM/L license number, batch code, and material grade stamp.

AUDIT STATUS BREAKDOWN & SCORING ALGORITHM

✓ **SATISFIED**

Declared material, test, or process matches published BIS specification. Evidence block cited.

✗ **CRITICAL GAP**

Statutory violation or missing mandatory test. Will trigger immediate BIS audit rejection or legal seizure.

! **NEEDS VERIFICATION**

Requires physical test certificate from NABL lab or calibration proof during physical audit.

Readiness Score (%) = (Satisfied Requirements / Total Applicable Requirements) × 100

45-SECOND DEMO WALKTHROUGH TO PRESENT TO JUDGES

- Select Preset:** Click "*Stainless Steel Water Bottles (IS 14543:2016)*". Notice fields auto-populate with Grade AISI 304 and ISO 9001:2015.
- Click "Run Comprehensive Compliance Audit":** In <800ms, the system evaluates the profile against the database.
- Show the Critical Gap Alert:** Point to the red banner: "*CRITICAL LEGAL GAP: Holding ISO 9001 does not satisfy Central QCOs. Mandatory ISI Mark (Scheme-I) required under Section 29 of BIS Act.*"
- Show Actionable Roadmap:** Demonstrate the step-by-step remediation plan directing the MSME to set up an in-house boiling acid test bath and apply on the Manakonline portal.

JUDGES' Q&A CHEAT SHEET

EXPECTED JUDGE QUESTION	PRECISE WINNING ANSWER
"Why not simply fine-tune GPT-4 on BIS PDFs?"	Fine-tuning bakes static knowledge into black-box weights without citations, cannot track weekly QCO amendments, and still hallucinates clauses. RAG with pgvector separates authoritative law from reasoning, guaranteeing 100% clause traceability.
"Is this legally binding?"	No, and Manak AI displays an explicit statutory disclaimer. It is an operational compliance intelligence tool that accelerates pre-audit readiness; official license grants remain the sovereign domain of BIS inspection officers.
"How does this scale to 19,000 standards?"	Our Supabase PostgreSQL pgvector engine with HNSW indexing scales sub-linearly. Retrieval latency remains <200ms whether querying 50 standards or 19,000 standards. Ingestion is fully automated via our chunking pipeline.